

CMBI Project: Quantification of Hepatic Iron Overload in Thalassemia Patients using MRI

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1. Thalassemia

Thalassemia is a genetic disorder causing the body to produce insufficient haemoglobin and red blood cells. The repeated transfusions by chronic blood transfusion cause progressive accumulation of iron in different body organs. Patients must undergo iron chelation therapy, otherwise affected patients die from endocrine and cardiac dysfunction in the second decade of life [1].

2. Iron overload

Hepatic iron overload may cause poor life quality and even lead to mortality. Hence, liver iron quantification is important for such patients. Around 70% of total body iron is deposited in the liver and hence is the main iron storage site [1-2]. Therefore, it was found that the iron content in the liver correlates closely with total body iron and accounts. Thus, quantifying liver iron content is an indicative of total body iron to guide, monitor, and initiate therapy. Hepatic iron content also serves as a biomarker for endocrine and cardiovascular complications in patients with thalassemia. Liver biopsy is the currently the gold standard technique for quantifying iron content in the liver. However, due to the technique's invasiveness, high cost, and the sampling error, it limits its widespread use.

3. MRI for iron overload estimation

Recently, magnetic resonance imaging (MRI) has been considered for evaluating iron content using T_2^* measurement [3] and has shown promising results. By plotting the signal intensity of a number of T_2^* -weighted images at different time to echo (TE's), against TE, the T_2^* (measured in ms), and its inverse $r = 1/T_2^*$, can be calculated using curve fitting.

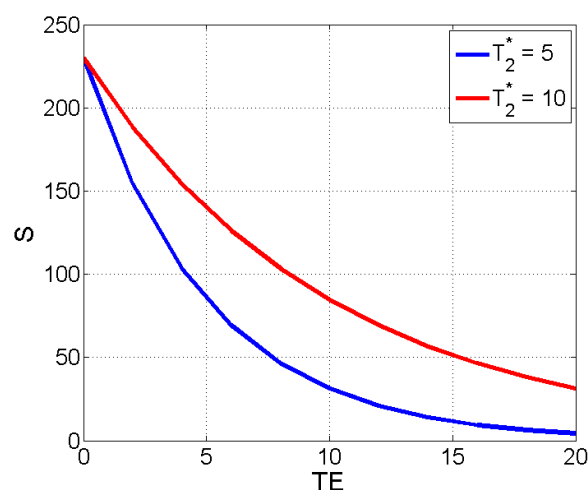


Fig. 1: Plot of time to echo (TE) versus corresponding intensity values for $T_2^* = 5$ and $T_2^* = 10$.

Figure 1 shows an example of two decay curves of different T_2^* values akin to those obtained using T_2^* weighted images. As the time to echo (TE) increases, the intensity follows an exponential decaying model with a time constant T_2^* . The higher the iron concentration is, the faster the decay of the intensities when TE increases, and hence the lower the corresponding T_2^* value. The core idea is to estimate the time constant T_2^* , which is an indicative to the total iron content in the liver. However, with the presence of noise and vasculature, the estimation of the T_2^* value, and hence the iron concentration, becomes difficult.

4. Fitting Models

The simplest model for data fitting is the single-exponential model given by,

$$s = a \exp(-rt), \quad (1)$$

where s is measured signal intensity, a is the signal intensity expected at $TE = 0$, and $t = TE$ is time to echo vector. The limitation of this model is that it assumes uniform iron concentration within each tissue voxel.

Numerous data-fitting models have been proposed to address the limitations of the single-exponential model such as the shifted single-exponential model. In this model, a constant c is added to account for long $T2^*$ components. This model can be written as,

$$s = a \exp(-rt) + c. \quad (2)$$

Moreover, the bi-exponential model has been introduced. In this model, two components are considered in the modelling process: an iron-dense, short- $T2^*$ component and an iron-sparse, long- $T2^*$ component). This model can be written as,

$$s = a \exp(-r_1 t) + b \exp(-r_2 t), \quad (3)$$

where a and b are the iron-dense, short- $T2^*$ and iron sparse, long- $T2^*$ components, respectively, and r_1 and r_2 are their corresponding inverse $T2^*$.

In this work, we will develop a robust, yet fast general framework for $T2^*$ estimation using $T2^*$ weighted magnetic resonance images, using the single-exponential model.

5. Patient datasets

In-vivo scans of the abdomen of four human subjects with sickle cell disease (age: 38 ± 12 years) were imaged on a 3Tesla Siemens Skyra MRI scanner (Siemens Healthcare, Erlangen, Germany) using a body surface coil and an 8-echo gradient echo (GRE) sequence with TEs ranging from 1 to 16.4ms in equal increments. The imaging parameters were as follows: repetition time (TR) = 5200 msec, matrix = 192×256 ; bandwidth = 1776 Hz/pixel, flip angle = 20, slice thickness = 10 mm, and field of view that depended on the patient size. A mid-liver axial slice was acquired in a single end expiration breath-hold. Figure 2 shows a sequence of $T2^*$ weighted images of the abdomen of a patient included in this study, with the parameters mentioned above. We can observe the decay of signal intensity in the liver as TE increases. Due to the heterogeneous iron concentration in the liver, susceptibility artifacts, and inclusion of vasculature, $T2^*$ measurements can be different in different parts of the liver. In this study, reference standard biopsy samples from the patients indicating iron overload content were not acquired, as this was not part of standard-of-care procedures. However, the images were analysed using the $T2^*$ mapping software on the scanner (which uses the single-exponential model) for $T2^*$ calculation. A circular region of interest (ROI) of about 4cm^2 was selected at the center of the vials' cross-sections and inside the liver's right lobe away from vasculature.

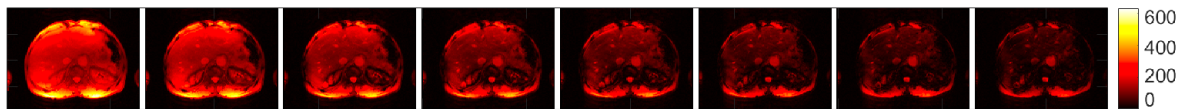


Fig.2: A sequence of liver images of different time to echo, of a patient with mild iron overload.

6. Core Tasks

Literature review

Task 1 – Review the literature on hepatic iron overload quantification using MRI. What is the most used fitting algorithm for estimation of $T2^*$?

Model derivation and implementation

Task 2 – Fit an exponential model

In this work, we will focus on the single exponential model. The aim is to estimate both a and r in Eq. (1). In order to do so, we rewrite Eq. (1) as the following unconstrained minimization problem

$$\underset{a_n, r_n}{\text{minimize}} \sum_{n=1}^N \frac{1}{2} \|s_n - a_n \exp(-r_n t)\|_2^2 + \alpha \Phi_1(a) + \beta \Phi_2(r). \quad (4)$$

Assign a total variation (TV) regularization to both a and r ; e.g., $\Phi_1(a) = \text{TV}(a)$ and $\Phi_2(r) = \text{TV}(r)$.

$$\underset{a_n, r_n}{\text{minimize}} \sum_{n=1}^N \frac{1}{2} \|s_n - a_n \exp(-r_n t)\|_2^2 + \alpha \text{TV}(a) + \beta \text{TV}(r) \quad (5)$$

The optimization problem in equation (5), although convex, cannot be solved using standard gradient-based methods due to the non-smooth terms. The core idea is to convert this unconstrained minimization problem into another constrained one by the application of a variable splitting operation. Finally, the obtained constrained problems are solved with using the alternating direction method of multipliers (ADMM) [6].

Solve the optimization problem in Eq. (5) using the alternating direction method of multipliers (ADMM) in order to estimate the unknown model parameters a and r :

- Assign suitable splitting parameters.
- Write down the corresponding Augmented Lagrangian to the optimization problem in (a).
- Write down the optimization problem to solve for each unknown variable in (b).
- Compute the gradient of each optimization function in (c).
- Write down the final optimization algorithm.

Task 3 – Implement the exponential fitting model in Task 2

Given the matlab functions provided in folder (matlab_files), insert the gradient equations computed in Task 2-d in relaxationEstimation.m. After doing so, using the main.m function provided, make sure it runs without errors.

Model validation using synthetic datasets

Task 4 – Create synthetic datasets

Given the matlab functions provided in the folder (matlab_files), create a sequence of 12 images of different time to echo using the function syntheticData.m as shown in Fig. (3). In a single plot, plot TE versus signal mean of intensities in each square. Comment on the result.

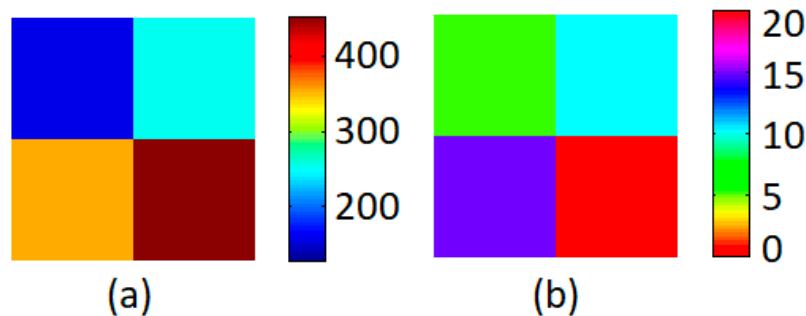


Fig. 3: Synthetic dataset (a) Image of 64x64 of different initial intensity values of $a = [155, 255; 355, 455]$ and (b) the corresponding $T2^*$ values $T2^* = [5, 10; 15, 20]$.

Task 5 – Test developed model

- Now, run the model we have developed in Task 2 using the synthetic data created in Task 4 and report estimated images of both a and $T2^* = 1/r$.
- Compute mean of a and $T2^* = 1/r$ in each box and compare with ground truth values.

- c) Add Gaussian noise of zero mean and a variance of 10 to the synthetic data. Repeat steps a) and b). Comment on the result.

Task 6 – Choice of regularization functions

In this work, we assigned total variation regularization functions to the unknown model parameter. What effect do this selection have on the results?

Results using patient datasets

Task 7 – Load and plot the MRI volumes

Given the scans of the four patients provided in folder (Patient Datasets), load a sequence of MRI (e.g. using matlab dicomread function) of Patient 1_2. Make sure the images have been loaded as double-precision arrays (if not, convert them). What is the dimensionality of the data? Plot the sequence of the images. Using the dicominfo function, report the time to echo (TE) of the data.

Task 8 – Plot TE versus signal intensity

Now, in a single plot, using a circular region of interest (ROI) of size 200 pixels squares approximately, at the ROI indicated in Fig. 4, plot TE versus mean of signal intensity value in the ROI for both Patients 1_2 and 3_8. Comment on the results.

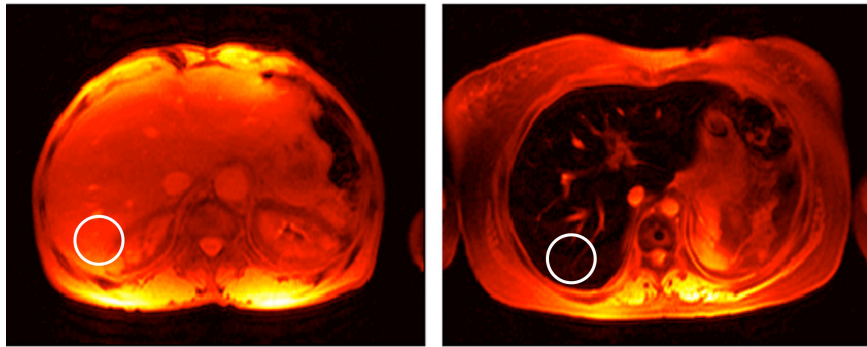


Fig. 4: Locations of different ROIs used for calculating $T2^*$ in patients with light (a) and heavy (b) iron overload.

Task 9 – Test developed model

Run the algorithm we have developed in Task 2 for each patient in folder (Patient Datasets) and report estimated images of both a and $T2^* = 1/r$. Comment on the results.

Using a circular region of interest (ROI) of size 200 pixels squares approximately at the ROI indicated in Fig. 2, compute mean $T2^*$ value for each patient and compare with MRI console estimates provided in the folder (Patient Datasets). Comment on the results.

7. Advanced Tasks

Now you will fit some more models to the data. These tasks are more open ended than the core tasks.

Task 10 – Fitting other models

The Levenberg Marquardt fitting algorithm (also called damped least-squares) is one of the widely used algorithms in the literature for $T2^*$ estimation. Briefly explain how it works, and the difference with the model you have developed in this work.

Task 11 – Fitting another model

Use the Levenberg Marquardt fitting algorithm to fit an exponential decay model as that in Eq. (1) using built-in matlab function, and repeat Tasks 5 and 9. Compare the results with those obtained using the fitting model you have developed.

References

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